Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU
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LANDSLIDE AND FLOOD THREAT ADVISORIES

To: Brgy. Chairperson
Brgy. Industrial Valley Complex
City of Marikina
Metropolitan Manila

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall **Olandes** **Bougainvillea Ext.** **Cinco Hermanos** **Montevista Gate 2**	N 14° 37' 42.6" E 121° 04' 42.3" N 14° 37' 27.3" E 121° 04' 44.4" N 14° 37' 17.6" E 121° 04' 37.9" N 14° 37' 33.6" E 121° 04' 37.4" N 14° 37' 46.9" E 121° 04' 41.6"	Very High Very High Very High High High	Brgy. Industrial Valley Complex has a generally progressive terrain where the eastern portion of barangay remain part of the Marikina River floodplains and hence, relatively flat. It then transgresses to sloping (moderately to steep) terrain towards the west. Flooding in the low elevated areas are of critical height due to the influx of rainfall runoff from higher elevated areas, and the overflow of the Marikina River. The barangay hall is situated at the base of the footslopes of the elevated portion of the barangay and neighboring Quezon City, which receives the surface runoff from the areas. Flooding in the area is compounded by its proximity to the Marikina River. During Ondoy, the 2nd floor was reached, while only 1st floor for Ulysses. The area of the barangay most proximal to Marikina River is the settlement. Observations points in *Olandes* and *Bougainvillea* revealed maximum flood heights exceeding 2m and almost reaching the 3rd floor of houses, during Ondoy. Meanwhile, during Ulysses, heights still exceeded 2m but ranges from top of 1st floor to half of 2nd floor, according to anecdotal accounts of residents. In Cinco Hermanos Subdivision, the subdivision remains affected by flooding despite being generally gently sloping, the low-lying areas, in the southeast. One resident along Pres. Quezon St. noted flood heights of around greater than 1m during Ondoy.

Remarks / Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system**. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Landslide Susceptibility Rating	Geohazards Assessment
Montevista Covered Court	N 14° 37' 51.3" E 121° 04' 31.2"	Very High	The western portion of Brgy. Industrial Valley Complex is situated in relatively sloping terrain (moderately to steep). There are isolated steep ridges that was probably created from human intervention primarily for land design (i.e. cut slope). These are susceptible to mass wasting, particularly rock fall, due to their sheer steepness. The most notable is in Montevista Subdivision. North of the subdivision's covered court, and perpendicular to a parking lot, is a steep slope of adobe where 2 incidents of rock fall was reported (2017 and 2019).

			The 2017 incident was previously assessed by a team from the MGB CO, however, the latter was not. The rock fall incident in 2017 was caused by rock erosion when fractures on the exposed adobe allowed rainfall runoff to percolate underneath causing weakening of the rock body and eventually slope failure. Slope mitigation measures had been installed since then as observed during the recent field assessment. Nevertheless, the 2017 and 2019 incidents caused damages to structures at its footslope. In addition, the exposed bedrock was observed to have cracks/fractures and the slope protection cement "sealant" is already hollow and may not effectively hold the slope. Structures observed atop the slope, which is within the jurisdiction of Quezon City, are also affected. In another part of the barangay, a basketball court is bounded to the north by an unmitigated slope with moderate gradient. Atop said slope is one of the main streets (Maj. Santos Dizon). Cracks, with vertical displacement, were observed along the asphalt and the paved sidewalk. One resident noted that these cracks along this portion of the road has been manifesting since before 2000, however, it was revealed that the road was rehabilitated recently. No landslide was recalled by the barangay official.

Remarks / Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
- ✓ **Observe for saturated ground and/or displaced roads surfaces** and report to the MGB/local authorities.
- ✓ **Maintenance and/or implementation of appropriate slope stabilization** (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended.
- ✓ Avoid settlement along the footslope, edge of the cliff/ravine, steep slope and ridge of hill/mountain.
- ✓ Barangay Disaster Risk Reduction Management Team must **implement monitoring and pre-emptive evacuation** during monsoon and typhoon season where extreme heavy rainfall occurs.
- ✓ **Activate** Barangay Disaster Risk Reduction Management Team in cooperation with the City Disaster Risk Reduction Management Team during monsoon and typhoon season.

- ✓ The CDRRMO together with the BDRRMO should conduct information and education awareness on this particular type of geohazard, particularly on landslide hazard to the landslide-prone communities.

Prepared by:

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Received by:

Designation:

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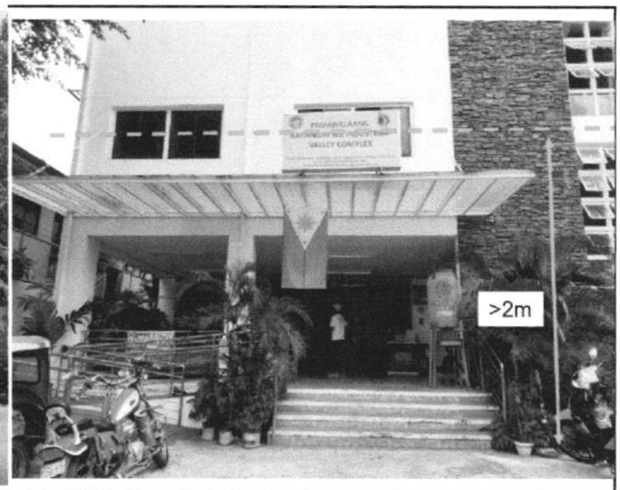
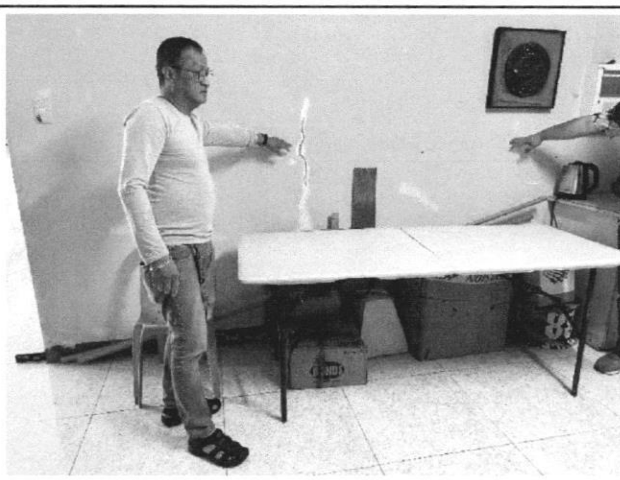
PHOTOS

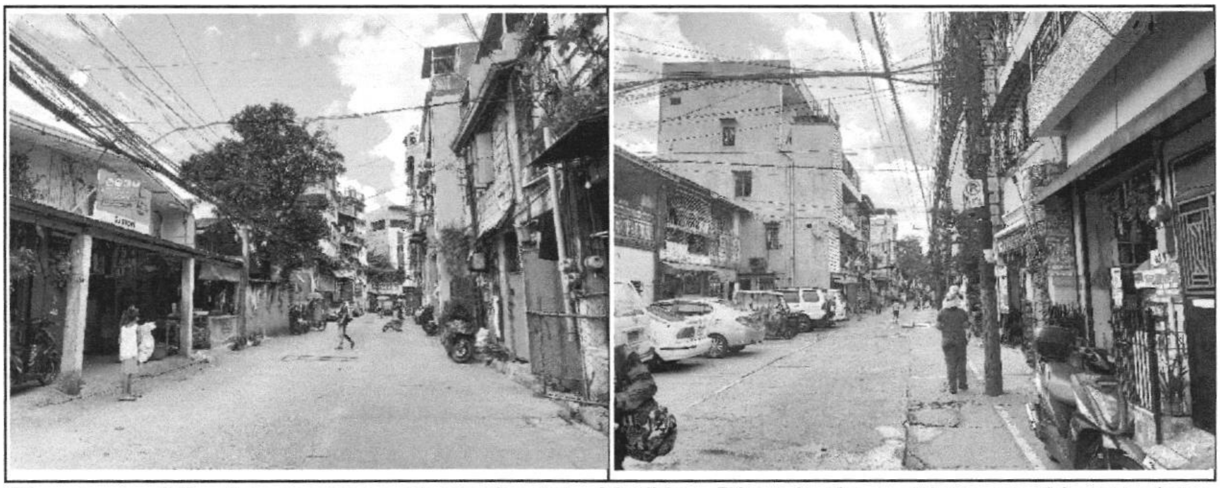




The barangay hall is situated at the base of the footslopes of the elevated portion of the barangay and neighboring Quezon City, which receives the surface runoff from the areas. Flooding in the area is compounded by its proximity to the Marikina River. During Ondoy, the 2nd floor was reached, while only 1st floor for Ulysses. GPS coordinates: N 14° 37' 42.6", E 121° 04' 42.3")

Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system**. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.



The area of the barangay most proximal to Marikina River is the settlement. Observations points in *Olandes* (left) and *Bougainvillea* (right) revealed maximum flood heights exceeding 2m and almost reaching the 3rd floor of houses, during Ondoy. Meanwhile, during Ulysses, heights still exceeded 2m but ranges from top of 1st floor to half of 2nd floor, according to anecdotal accounts of residents. (GPS coordinates: N 14° 37' 27.3", E 121° 04' 44.4"; N 14° 37' 17.6", E 121° 04' 37.9")

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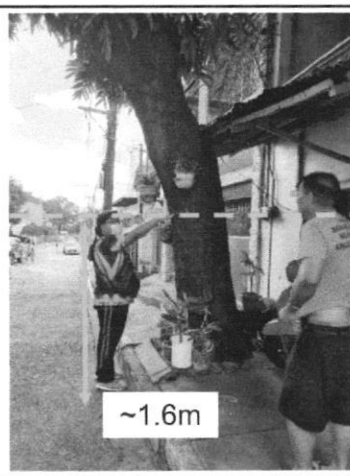

Although the subdivision is generally gently sloping, the low-lying areas, in the southeast, are still affected by flooding. One resident along Pres. Quezon St. noted flood heights of around greater than 1m during Ondoy. (GPS coordinates: N 14° 37' 33.6", E 121° 04' 37.4")

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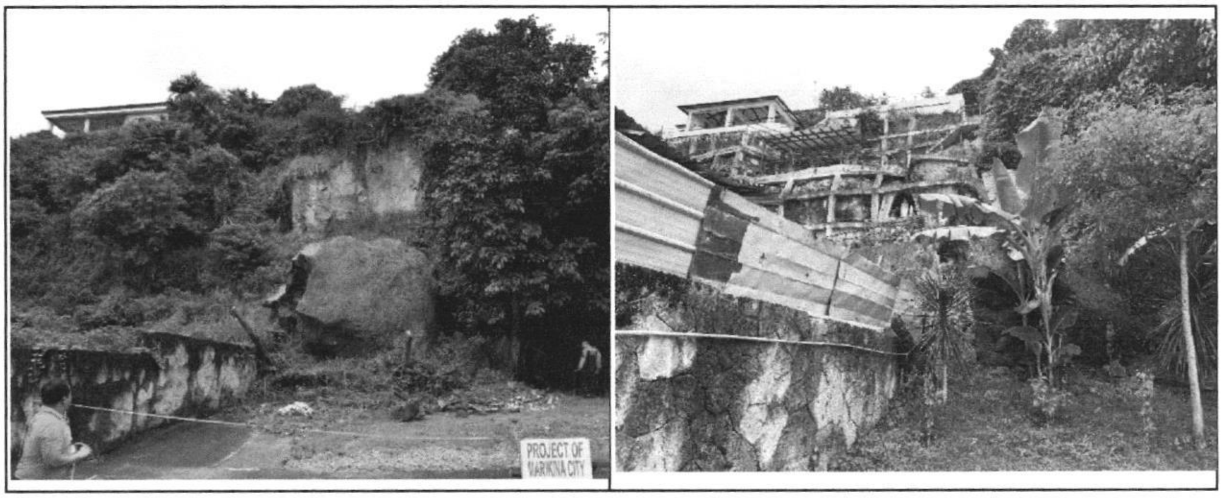

The upper photos show the area in Montevista Subdivision in Brgy. Industrial Valley Complex where landslides occurred in 2017 and 2019. The left photo shows the 2017 landslide incident while the upper right photo shows the current state of the area in 2023. The bottom photos show the exposed bedrock was observed to have cracks/fractures and the slope protection cement "sealant" is already hollow and may not effectively hold the slope. A structure was also observed atop the slope, which is within the jurisdiction of Quezon City. (GPS coordinates: 14° 37' 51.3" N, 121° 04' 31.2" E)

Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
- ✓ **Observe for saturated ground and/or displaced roads surfaces** and report to the MGB/local authorities.
- ✓ **Maintenance and/or implementation of appropriate slope stabilization** (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended.
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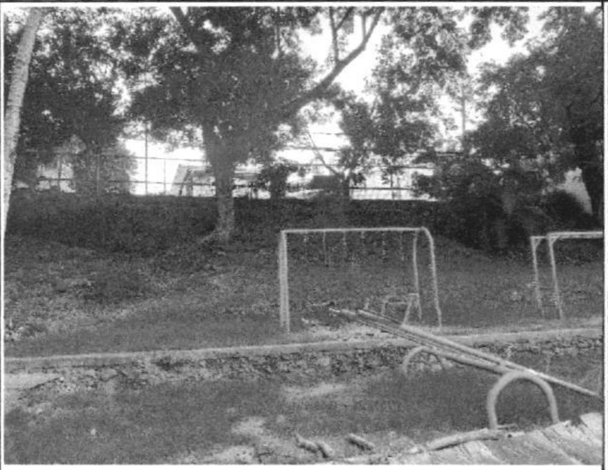


Northwest of the barangay, the IVC court is bounded to the north by an unmitigated slope with moderate gradient. Atop said slope is one of the main streets (Maj. Santos Dizon). Cracks, with vertical displacement, were observed along the asphalt and the paved sidewalk. One resident noted that these cracks along this portion of the road has been manifesting since before 2000, however, it was revealed that the road was rehabilitated recently. No landslide was recalled by the barangay official.

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